



KrishiVision AI Report

AI-Powered Satellite Crop Monitoring & Geographic Analytics

CROP Cotton	CULTIVATED AREA 221.16 Acres
DISTRICT Dharwad	STATE Karnataka
HEALTH STATUS Poor	HEALTH SCORE 23 / 100
CURRENT GROWTH STAGE Planting	ESTIMATED HARVEST In 30 Days

EXECUTIVE SUMMARY

The monitored Cotton crop in Dharwad district (Karnataka) currently shows Poor vegetation health based on the latest available satellite observations. The current growth stage is Planting, with an estimated harvest window of In 30 Days. The calculated agricultural health score is 23/100, derived dynamically via Google Earth Engine and Sentinel-2 index pipelines.

DATA SOURCES & TELEMETRY INTEGRATION

Sentinel-2 L2A Multispectral Imagery | NDVI Vegetation Density Indexes | EVI Enhanced Canopy Activity | NDWI Soil Moisture Telemetry | Open-Meteo Weather Grids | KrishiVision AI Baseline Reference Models

SATELLITE INTELLIGENCE

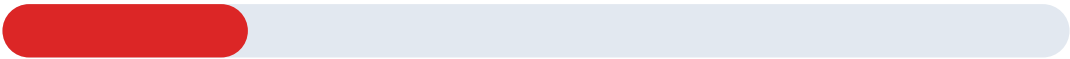
SATELLITE DATA UNAVAILABLE

No valid satellite observation is currently available for this crop. Please configure active farm boundary coordinates in the map settings to trigger automatic Sentinel-2 diagnostics.

OBSERVATION DATE 2026-08-23	SATELLITE PLATFORM Sentinel-2 L2A
RESOLUTION / CORRECTION 10m / Sen2Cor BOA	NDVI (VEG DENSITY) 0.0196
EVI (CANOPY ACTIVITY) 0.0591	NDWI (MOISTURE INDEX) 27.3000
LOCAL TEMPERATURE 26.8°C	CLOUD COVERAGE 12% Limit

CROP HEALTH ANALYSIS

HEALTH SCORE: 23 / 100



POOR

NDVI STATUS Vegetation stress detected	EVI STATUS Stable Photosynthesis
MOISTURE NDWI Healthy canopy condition	OBSERVATION QUALITY Excellent (Cloud < 15%)

HEALTH INTERPRETATION

- Active Chlorophyll Density: The NDVI score of 0.0196 indicates vegetation stress detected.
- Canopy Structure: EVI of 0.0591 points to stable photosynthetic activity and leaf density growth.
- Water Content: NDWI of 27.3000 suggests healthy canopy condition in the monitored quadrants.

FIELD AREA METRICS

Metric Parameter	Area Coverage	Percentage
Total Field Area	221.16 Acres	100.0%
Healthy Area (High Vigor)	154.81 Acres	70%
Stressed / Affected Area	66.35 Acres	30%
Bare Soil / Roads / Other	11.06 Acres	5.0%

GROWTH & HARVEST INTELLIGENCE



CURRENT STAGE Planting	EXPECTED NEXT STAGE Maturity
STAGE PROGRESS 15%	CONFIDENCE INDEX 95.0%

HARVEST PREDICTION

ESTIMATED HARVEST TIME 30 DAYS	Expected Date: 26 September 2026 Prediction Confidence: 95.0%
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AI RECOMMENDATIONS & FIELD INSIGHTS

- Moisture Management: Critical moisture stress detected! Immediate crop irrigation required.
- Nutrient Schedule: High vegetative distress. Apply targeted micronutrient foliage spray.
- Pathogen Alert: Inspect and isolate crop leaf nodes for rust or blight infections.

CROP REFERENCE INFORMATION

Crop Name	Cotton	Crop Category	Commercial
Scientific Name	Gossypium hirsutum	Suitable Soil	Black cotton soil (Regur)
Water Requirement	Moderate (500 - 1100 mm)	Suitable Climate	Warm and Dry (21°C - 30°C)
Typical Duration	160 - 180 Days	Growing Season	Kharif

REPORT VERIFICATION

REPORT VALIDATION & SECURITY

ID: KV-DHA-COT-202608271047

Timestamp: 2026-08-27 10:47

Data Sources: Sentinel-2 L2A, Open-Meteo, AgroMonitoring

Scan to verify authentic classification parameters.

